

Model-Agnostic Explanations with Partial Dependence Plots

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Abstract

This report applies model-agnostic explanation methods to two regression problems: bike rental demand prediction and house price prediction. In both cases, a random forest model is used as a predictive approximation, and partial dependence plots are used to interpret how selected input features influence the model predictions. For the bike rental dataset, we analyze the influence of time, temperature, humidity, and wind speed on predicted bike counts. We also study the combined effect of temperature and humidity using a two-dimensional partial dependence plot. For the house price dataset, we analyze how structural features such as bedrooms, bathrooms, living area, and floors affect predicted house prices.

1 Introduction

Complex machine learning models can provide accurate predictions, but their internal decision process is often difficult to interpret directly. Model-agnostic explanation methods help address this problem by explaining the behavior of a trained model without depending on its internal structure.

In this report, we use partial dependence plots (PDPs) to analyze the behavior of random forest regression models. A partial dependence plot shows how the predicted outcome changes when one input feature varies, while the remaining features are averaged over the observed data. This makes PDPs useful for understanding the marginal effect of individual variables on model predictions.

The analysis is divided into two practical use cases. First, we study daily bike rental demand using weather, seasonal, and temporal features. Second, we analyze house price predictions using structural property features.

2 Bike Rental Demand Analysis

2.1 Data and Model

The first analysis uses the Bike Sharing dataset, where the target variable is `cnt`, the total number of rented bikes per day. The input features include weather-related variables, seasonal indicators, and a time trend variable.

A random forest regressor was trained to predict daily bike rental counts. Since random forests are ensemble models and are not directly interpretable from a single equation, partial dependence plots are used to explain how the model predictions react to changes in selected features.

2.2 One-Dimensional Partial Dependence Analysis

The first set of PDPs analyzes the marginal effect of individual variables on the predicted number of rented bikes. The random forest model was trained using weather, seasonal, and temporal features. In this section, the interpretation focuses on these four variables: `days_since_2011`, `temp`, `hum`, and `windspeed`.

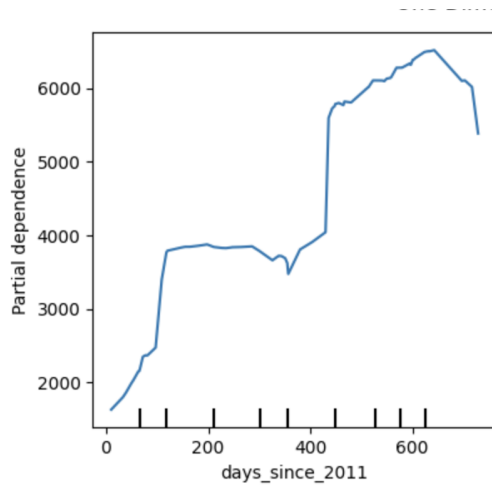


Figure 1: Partial dependence of predicted bike rentals on `days_since_2011`.

Figure 1 shows a strong positive time trend. At the beginning of the observation period, the predicted number of rentals is relatively low. The prediction increases over time, with a particularly clear jump in the later part of the dataset. This suggests that the random forest model learned an increasing demand pattern over the two-year period. From an operational perspective, this indicates that bike-sharing usage became more established over time, so later periods require higher expected bike availability.

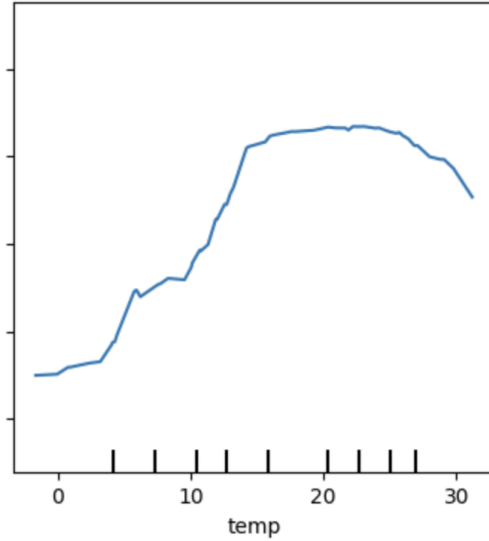


Figure 2: Partial dependence of predicted bike rentals on temperature.

Figure 2 shows that temperature has a strong positive effect on predicted bike rentals. Demand is low at very cold temperatures, then increases clearly as temperature rises. The strongest increase appears between approximately 5°C and 15°C. After that, the effect becomes flatter, with high predicted demand under mild to warm conditions. At very high temperatures, the predicted demand slightly decreases, which may indicate that extremely warm conditions become less comfortable for cycling.

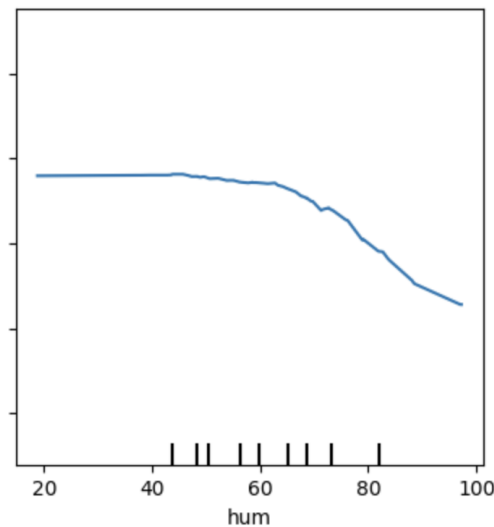


Figure 3: Partial dependence of predicted bike rentals on humidity.

Figure 3 indicates a negative relationship between humidity and predicted bike rentals. For lower and medium humidity values, the predicted demand remains relatively stable. However, from higher humidity levels onward, the predicted number of rentals decreases.

more clearly. This suggests that the model associates humid conditions with lower cycling demand, probably because high humidity makes outdoor activity less comfortable.

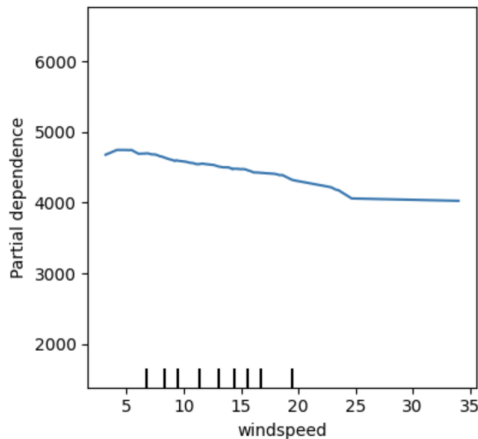


Figure 4: Partial dependence of predicted bike rentals on wind speed.

Figure 4 shows a slightly negative effect of wind speed. Predicted rentals are higher at low wind speeds and gradually decrease as wind speed increases. The decrease is less steep than for humidity or temperature, but the direction is still clear. This suggests that windy conditions reduce the attractiveness of cycling, although wind speed is not as dominant as temperature or the time trend.

Overall, the one-dimensional PDPs show that bike rental demand is mainly driven by a positive time trend and by weather comfort. Warmer temperatures increase predicted demand, while high humidity and stronger wind reduce it. These results are consistent with practical expectations: bike rentals are more attractive under mild, comfortable weather conditions and became more frequent over the observed time period.

2.3 Two-Dimensional Partial Dependence Analysis

To analyze the combined effect of temperature and humidity, a two-dimensional partial dependence plot was generated. This plot shows how the predicted number of rented bikes changes when both variables vary at the same time, while the remaining model features are averaged over the sampled observations.

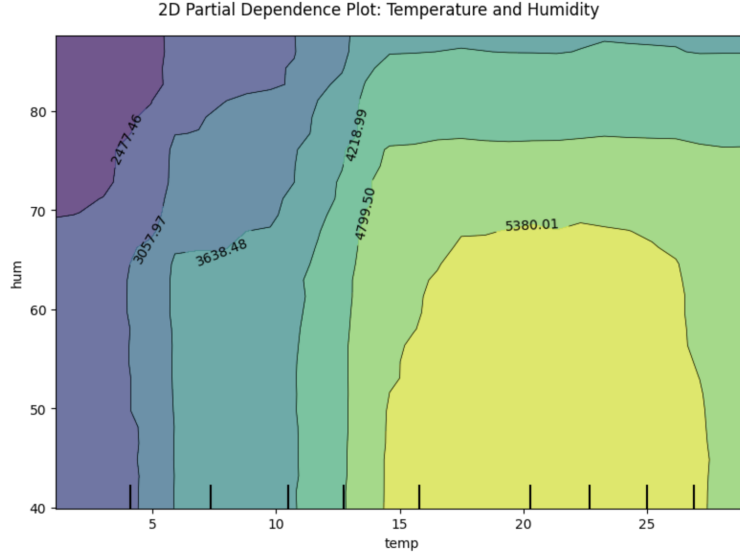


Figure 5: Two-dimensional partial dependence plot for temperature and humidity.

Figure 5 shows that predicted bike rental demand is lowest under cold and humid conditions. In the left part of the plot, where temperatures are low, the predicted number of rentals remains comparatively low across humidity levels. As temperature increases, predicted demand rises substantially.

The highest predicted demand appears for moderate to warm temperatures, roughly between 15°C and 27°C, combined with medium humidity levels. In this region, the model predicts more than 5,000 daily rentals. However, at very high humidity levels, predicted demand decreases again, even when temperature is relatively warm. This suggests that the model learned a comfort-related interaction: warm weather increases bike usage, but excessive humidity reduces the attractiveness of cycling.

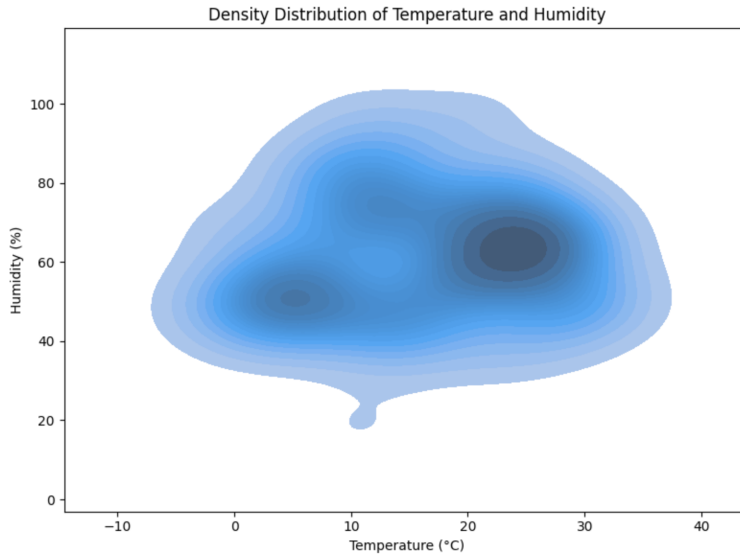


Figure 6: Observed density distribution of temperature and humidity in the bike rental data.

Figure 6 shows where temperature and humidity combinations occur most frequently in the observed data. The densest areas are mainly around mild to warm temperatures with medium humidity. This is important because PDP values are more reliable in regions where many observations exist. In sparse regions, such as very low temperatures or extreme humidity combinations, the model predictions should be interpreted more carefully.

Overall, the two-dimensional analysis confirms the findings from the one-dimensional PDPs. Bike rental demand is highest under warm and comfortable conditions, while cold temperatures and high humidity reduce predicted demand. From a planning perspective, this suggests that bike availability should be increased on warm, moderately humid days, while lower demand can be expected during cold or very humid conditions.

3 House Price Prediction Analysis

3.1 Data and Model

The second analysis uses the `kc_house_data.csv` dataset. The target variable is `price`, representing the sale price of a house. A random forest regressor was trained using the features `bedrooms`, `bathrooms`, `sqft_living`, `sqft_lot`, `floors`, and `yr_built`.

As in the bike rental analysis, partial dependence plots are used to explain how the trained random forest model reacts to changes in individual input variables. The following interpretation focuses on the four main variables: `bedrooms`, `bathrooms`, `sqft_living`, and `floors`.

3.2 Partial Dependence Analysis

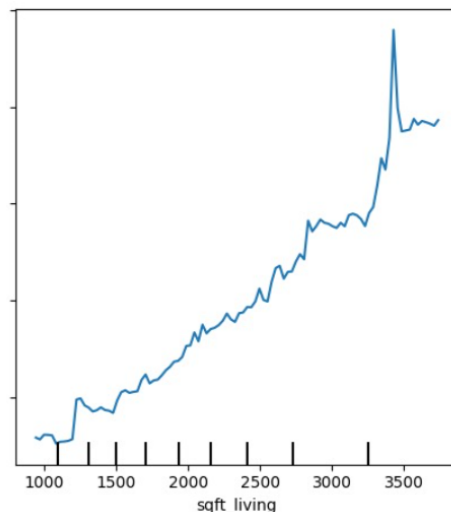


Figure 7: Partial dependence of predicted house price on `sqft_living`.

Figure 7 shows the clearest and strongest relationship among the analyzed house features. As `sqft_living` increases, the predicted house price rises almost continuously. This indicates

that the random forest model strongly associates larger living area with higher property value. The effect is intuitive, because living space directly reflects the usable indoor size of the house and is therefore one of the most important structural indicators of price.

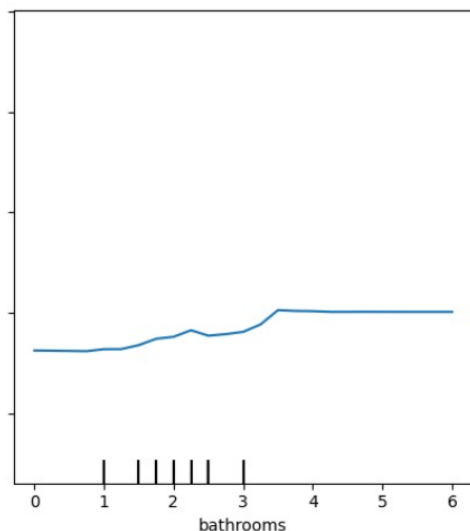


Figure 8: Partial dependence of predicted house price on **bathrooms**.

Figure 8 shows a generally positive relationship between the number of bathrooms and predicted house price. The model prediction increases gradually as the number of bathrooms rises. This suggests that houses with more bathrooms are typically associated with higher value. However, this effect should not be interpreted as purely causal, because the number of bathrooms is likely correlated with other property characteristics such as house size, quality, and overall comfort.

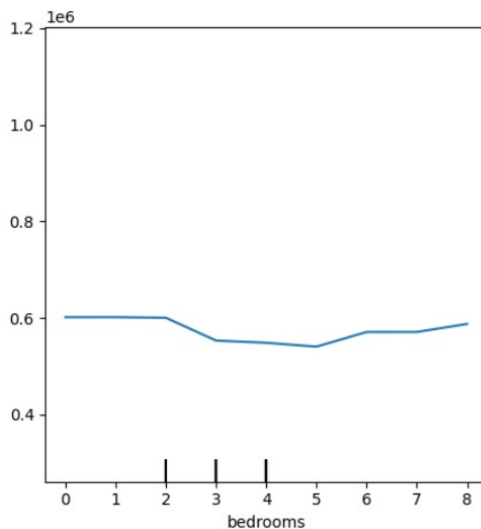


Figure 9: Partial dependence of predicted house price on **bedrooms**.

Figure 9 shows a less intuitive pattern. The predicted price does not increase with the

number of bedrooms. Instead, the curve decreases after around two bedrooms and then remains relatively flat. This suggests that, in the fitted model, bedroom count alone is not a strong positive driver of price. A possible explanation is that the number of bedrooms does not capture the size or quality of the rooms. For example, houses with many bedrooms may have smaller rooms or may not necessarily provide more total living space. Therefore, `sqft_living` appears to be a more meaningful size-related feature than `bedrooms`.

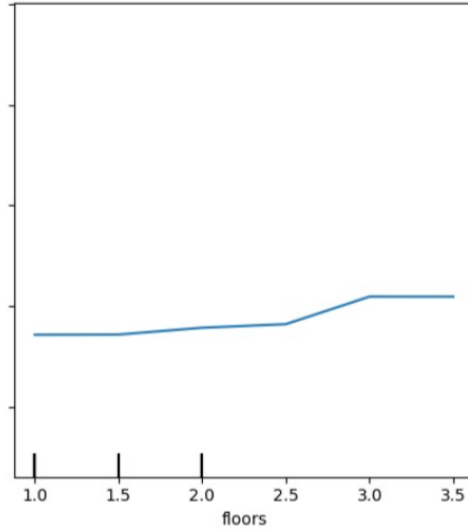


Figure 10: Partial dependence of predicted house price on `floors`.

Figure 10 indicates a moderate positive effect of the number of floors. Predicted prices increase slightly from one to around three floors and then stabilize. Compared with `sqft_living`, this effect is weaker. This suggests that the model considers the vertical structure of a house relevant, but less important than the total living area. In practical terms, having more floors may be associated with larger or more valuable homes, but the amount of usable living space remains the stronger predictor.

Overall, the house price PDPs show that the random forest model mainly relies on living area and structural comfort features when predicting house prices. Among the four analyzed variables, `sqft_living` has the strongest and most consistent positive influence. Bathrooms and floors also increase predicted price, but their effects are weaker and likely correlated with overall house size. Bedrooms show a non-linear and less directly positive relationship, highlighting that not all count-based features are equally informative for predicting property value.

4 Conclusion

This report used partial dependence plots to explain the behavior of random forest regression models in two different prediction scenarios. Instead of interpreting the internal structure of the random forest directly, PDPs allowed us to analyze how selected input variables influence the model predictions on average.

For the bike rental dataset, the results show that predicted demand is mainly driven by time and weather conditions. The variable `days_since_2011` indicates a clear positive time trend, suggesting that bike rental demand increased over the observed period. Temperature also has a strong positive effect: warmer days are associated with higher predicted rental counts. In contrast, high humidity and stronger wind speed reduce predicted demand, which is consistent with the idea that uncomfortable weather conditions make cycling less attractive.

The two-dimensional PDP for temperature and humidity confirms this interpretation. The highest predicted demand occurs under warm and moderately humid conditions, while cold temperatures and very high humidity lead to lower predicted bike counts. The density plot also shows that PDPs should be interpreted more carefully in sparse regions of the feature space, where only few observations are available.

For the house price dataset, `sqft_living` is the strongest and most consistent driver of predicted price. Larger living area clearly increases the model prediction. Bathrooms and floors also show positive effects, although weaker than the effect of living area. Bedrooms show a less direct relationship, indicating that the number of rooms alone is less informative than the total usable space.

Overall, PDPs proved useful for translating complex random forest models into understandable feature effects. However, the results should be interpreted as model explanations rather than causal relationships. The plots show what the trained model has learned from the data, but they do not prove that changing one feature would directly cause the predicted outcome to change in the same way.